

1- What is the primary reasons for the shift from schematic entry to Hardware Description Languages (HDLs)?

- ☒ A) HDLs allow for easier simulation of designs.
- B) HDLs are easier to learn than schematic entry.
- C) HDLs cannot handle complex designs.
- D) HDLs were more expensive to use.

2- Which Hardware Description Language was developed in the early 1980s for the American Department of Defense?

- A) Verilog
- B) SystemVerilog
- ☒ C) VHDL
- D) HDL++

3- VHDL was originally developed for what purpose?

- A) To design graphical schematics
- ☒ B) To model and simulate digital systems
- C) To design mechanical systems
- D) To optimize power consumption

4- Which of the following is true about schematic entry compared to HDLs?

- A) Schematic entry is easier to simulate.
- B) Schematic entry is more efficient for large designs.
- C) Schematic entry requires less effort for changes.
- ☒ D) Schematic entry can become annoying for complex designs.

5- What is the role of synthesis tools in relation to VHDL?

- A) They generate new languages for VHDL.
- B) They compile VHDL code.
- ☒ C) They create and optimize the physical implementation of designs.

6- What can HDLs describe in electronic circuits?

- A) Only the layout
- B) Only the physical size
- ☒ C) The circuit's operation and design but usually not its layout
- D) The operating voltage only

7- What are the two main HDLs available?

- A) SystemVerilog and SystemVHDL
- B) Catapult C and Impulse C
- ☒ C) Verilog and VHDL
- D) Handel-C and Catapult C

8- What does VHDL stand for?

- ☒ A) Very High Speed Integrated Circuit Hardware Description Language
- B) Virtual High Description Language
- C) Variable High Speed Digital Language
- D) Very High-Speed Description Language

9- What is one benefit of using HDLs for system-level design?

- ☒ A) They allow design of both hardware and software concurrently.
- B) They increase the time needed for the design process.
- C) They reduce the level of abstraction in the design.
- D) They limit the use of high-level languages.

10- Why are HDLs important in the design process?

- A) They increase the complexity of designs.
- B) They allow for precise layout of electronic circuits.
- ☒ C) They enable verification and simulation of a circuit's operation.

11- Why is it important to focus on a subset of VHDL constructs and operators?

- A) To make the code less readable.
- B) To avoid compatibility with other HDLs.
- ☒ C) To ensure the VHDL code can be synthesized into hardware.
- D) To create testbenches for verification purposes.

12- Which VHDL construct should generally be avoided in synthesizable VHDL?

- A) Case statements
- ☒ B) For loops with unknown bounds
- C) Simple arithmetic operators
- D) Use of signals and variables

13- What do signals in VHDL define?

- A) The operating frequency of the circuit
- ☒ B) The structure and behavior of the wires connecting circuit components
- C) The physical layout of the circuit components
- D) The memory allocation within the circuit

14- What is one of the primary uses of signals in VHDL?

- A) Managing user input and output devices.
- B) Serving as constants in the design.
- ☒ C) Controlling and managing data flow in the circuit.
- D) Optimizing memory usage in the circuit.

15- How are signals declared in VHDL?

- A) Using the VARIABLE keyword.
- ☒ B) Using the SIGNAL keyword.
- C) Using the CONST keyword.

16- What type of characters can be used in signal names in VHDL?

- A) Special characters such as #, \$, and %
- ☒ B) Only alphanumeric characters and underscores
- C) Any ASCII characters
- D) Only capital letters

17- What are some types that single-bit signals can have in synthesizable VHDL?

- A) int, float, and double
- ☒ B) STD_LOGIC_VECTOR, UNSIGNED, and SIGNED
- C) boolean, char, and string
- D) 0, 1, and Z

18- What types of multiple-wire signals are commonly used in synthesizable VHDL?

- A) char_array and string_array
- ☒ B) std_logic_vector, unsigned, and signed
- C) int_vector and float_vector
- D) boolean_array and bit_vector

19- What is the purpose of the STD_LOGIC_VECTOR type in VHDL?

- A) To perform arithmetic operations
- ☒ B) To represent a sequence of STD_LOGIC values
- C) To define floating-point numbers
- D) To handle programming language constructs

20- What additional information do UNSIGNED and SIGNED types carry?

- A) Information for handling floating-point arithmetic
- B) Information for character processing
- ☒ C) Information for numerical connotation and arithmetic operations

21- Why is the complexity of different signal types important in VHDL?

- A) To make the design process more complicated
- ☒ B) To provide precise control over circuit operations and arithmetic
- C) To introduce unpredictability into the design
- D) To increase the likelihood of design errors

22- What are some values allowed for STD_LOGIC types in VHDL?

- A) A, B, C, D
- B) true, false, maybe
- ☒ C) 1, 0, U, X
- D) high, low, medium

23- What is the function of the datapath (execution unit)?

- A) Controls data movement in the circuit.
- B) Manages memory allocation in the circuit.
- ☒ C) Manipulates and processes data.
- D) Determines the sequence of operations performed by the circuit.

24- Which of the following components can be part of the datapath?

- ☒ A) Registers, multiplexers, and decoders
- B) Program counter and instructions
- C) High-level programming constructs
- D) Memory controllers and processors

25- What is the main role of the controller (Control Unit)?

- A) To manage the power supply of the circuit
- ☒ B) To control data movement in the Datapath
- C) To perform arithmetic operations

26- Which of the following is a characteristic of a programmable controller?

- A) It remains fixed once designed.
- B) It uses a hardwired state machine.
- ☒ C) It uses a program counter to point to the next instruction.
- D) It performs only one specific task.

27- What does the controller provide to the Datapath?

- A) Data processing resources
- ☒ B) Signals to activate processing tasks
- C) Memory management
- D) Manual user input

28- What is one example of a programmable controller?

- ☒ A) A microprocessor
- B) A hardwired state machine
- C) A fixed-function circuit
- D) A combinational logic circuit

29- What does the controller do in terms of scheduling operations?

- A) It uses a hardwired state machine to manage user input.
- B) It provides signals to activate the power supply.
- ☒ C) It determines the sequence of operations performed by the Datapath.
- D) It selects signals to activate multiplexers in the Datapath.

30- What is the function of a Finite State Machine (FSM)?

- A) To store large amounts of data.
- ☒ B) To model a system that transits among a finite number of internal states.
- C) To design analog circuits.

31- What is the primary application of an FSM?

- A) To store instructions for future use.
- B) To control and manage power supply.
- ☒ C) To act as the controller for a large digital system.
- D) To handle user inputs and outputs.

32- What are the two main types of FSMs?

- A) Synchronous and asynchronous FSMs
- B) Linear and nonlinear FSMs
- ☒ C) Mealy and Moore FSMs
- D) Finite and infinite FSMs

33- How do the Mealy and Moore FSMs differ?

- A) Moore FSMs are faster than Mealy FSMs.
- B) Mealy FSMs use more states than Moore FSMs.
- ☒ C) In Mealy FSMs, output depends on the present state and inputs, while in Moore FSMs, output depends only on the present state.
- D) Moore FSMs are susceptible to input signal disturbances.

34- Why is a Moore FSM generally slower than a Mealy FSM?

- A) Moore FSMs require more states.
- ☒ B) The output of a Moore FSM is typically obtained one clock period later.
- C) Moore FSMs have direct paths from inputs to outputs.
- D) Moore FSMs have combinational paths between inputs and outputs.

35- What is one advantage of a Mealy FSM over a Moore FSM?

- ☒ A) It typically uses less states for a given application.
- B) It responds to inputs one clock cycle later than an equivalent Moore FSM.

36- Which type of FSM is less likely to affect the critical path of an entire circuit?

- A) Mealy FSM
- B) Both types have equal effect on the critical path
-  C) Moore FSM
- D) Neither type affects the critical path

37- How can an equivalent Mealy FSM be derived from a Moore FSM?

- A) By increasing the number of states in the Moore FSM.
- B) By combining multiple Moore FSMs.
-  C) By converting the outputs of the Moore FSM into combinational logic.
- D) By decreasing the number of transitions in the Moore FSM.